SIMATS ENGINEERING
ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1503

COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct
I **Sweta R (192511408)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: Sweta R

Assessment Tasks Concept Mapping Tasks
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

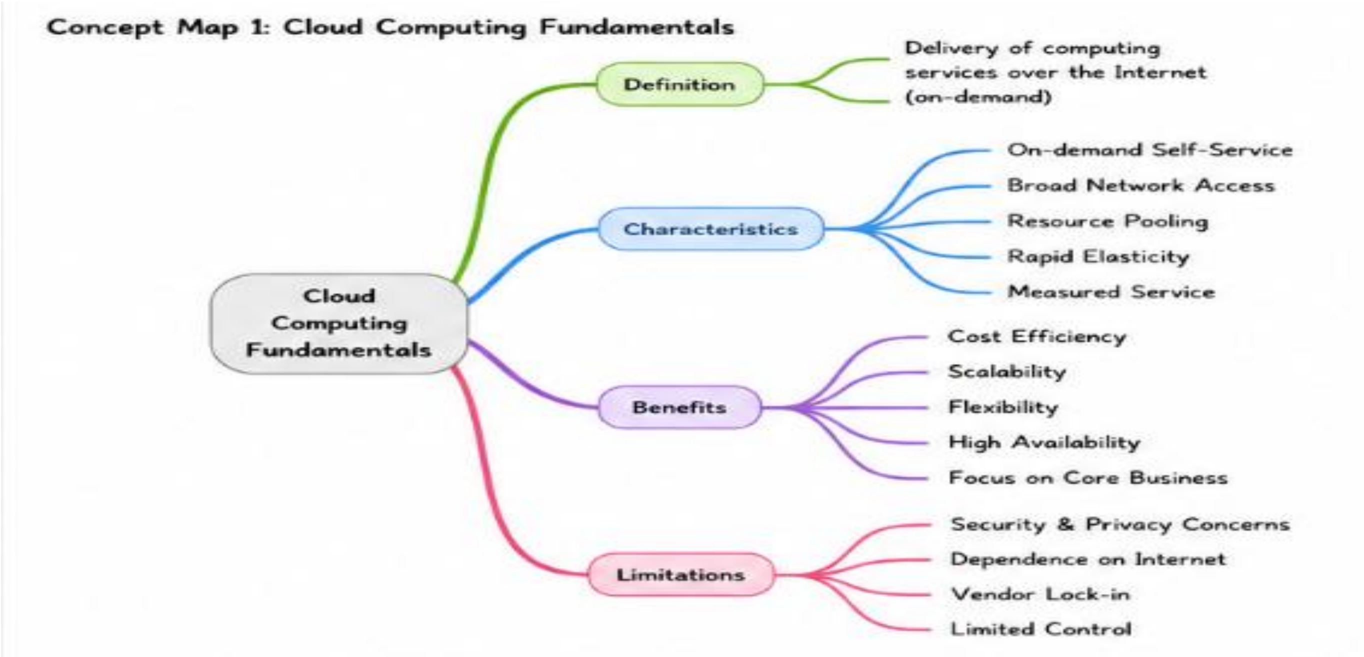
Expected concept nodes:
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
		accurate and meaningful			
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

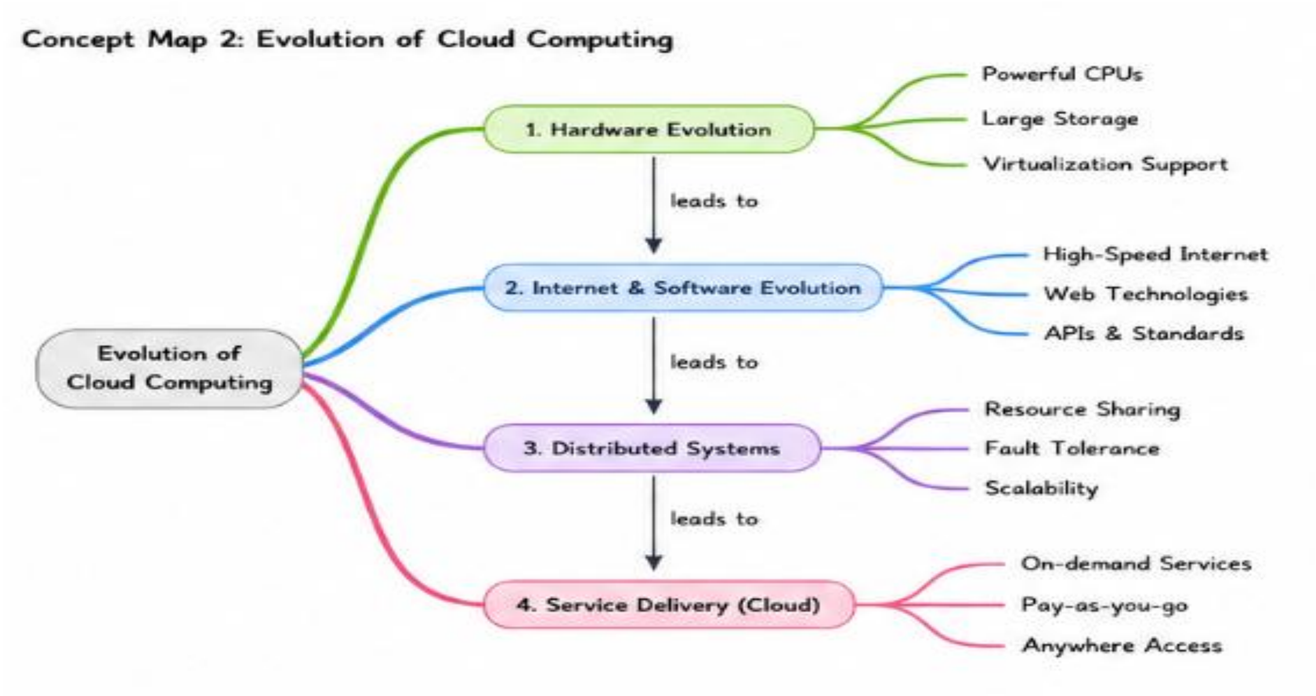
Concept Map 1

Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitation



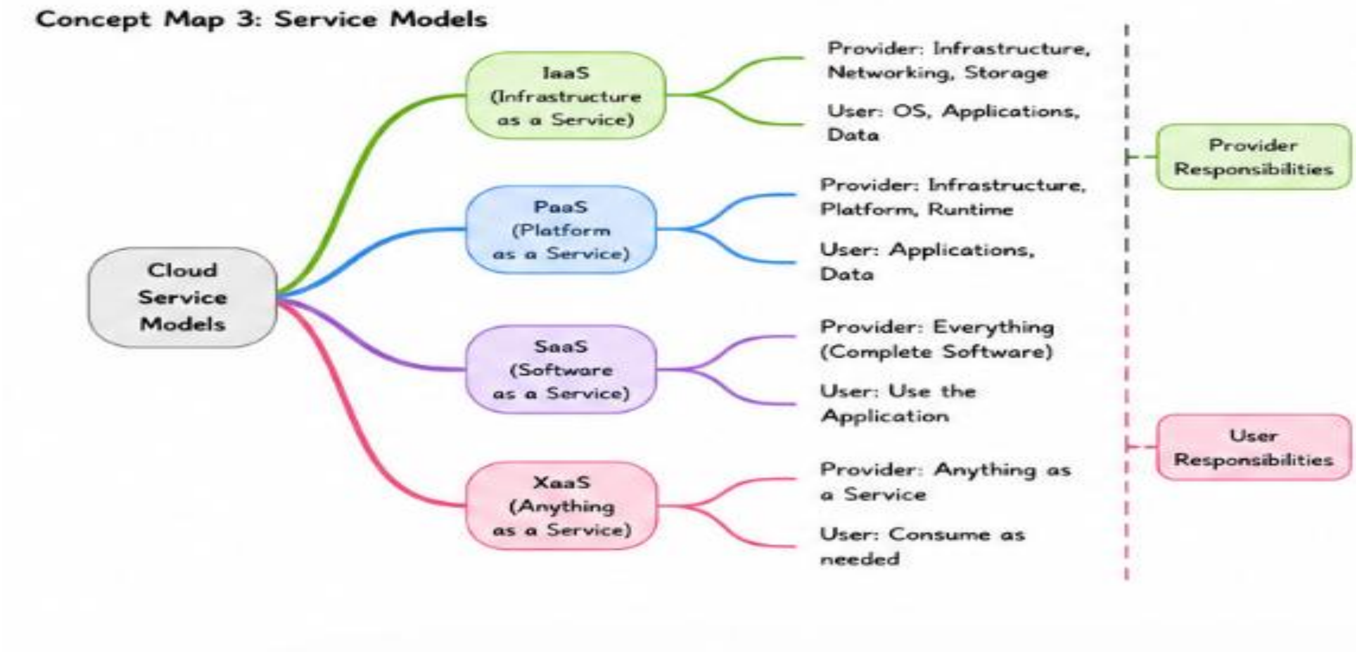
Concept Map 2

Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



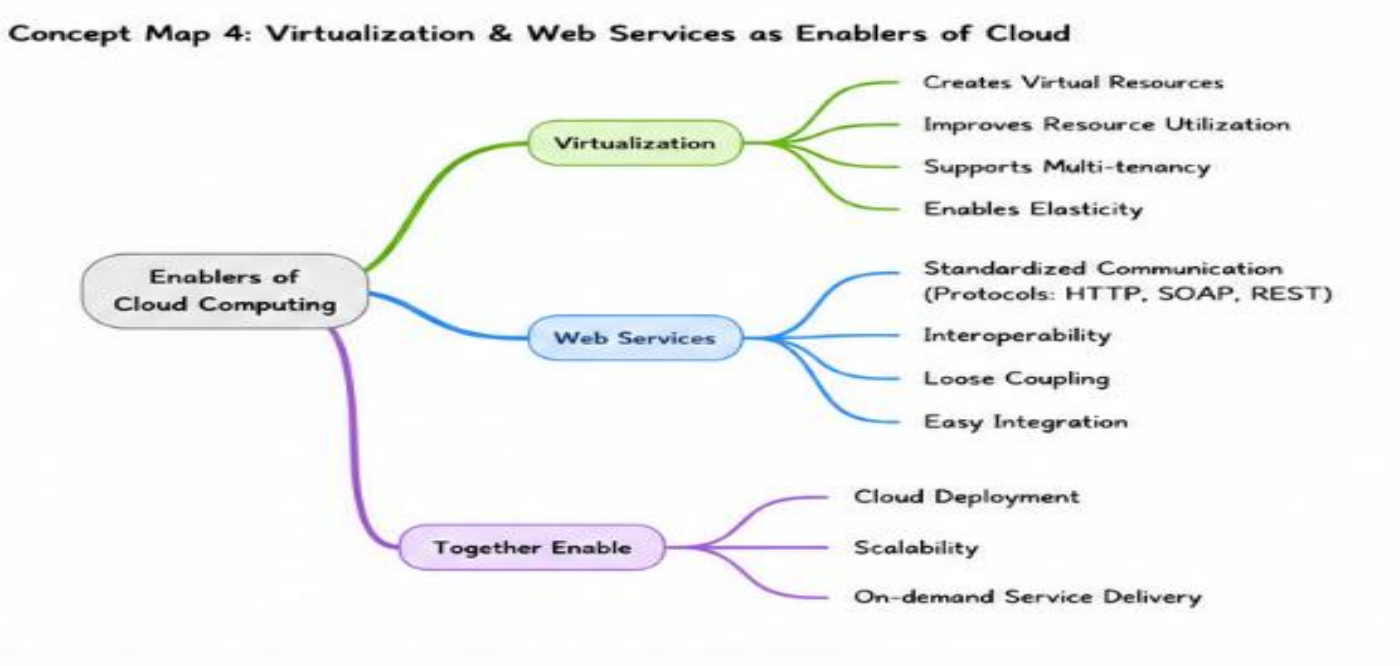
Concept Map 3

Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Concept Map 4

Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



1. **Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?**

Effectiveness

- Presents cloud computing topics in a simple and structured manner.
- Highlights how different cloud technologies are connected.
- Breaks down complex ideas into smaller, easy-to-follow concepts.
- Enables quick understanding and revision through a visual layout**.**

Design Decisions

- Chose the main topic as the starting point of each concept map.
- Linked related ideas with branches to show their relationships.
- Grouped similar concepts together to avoid confusion.
- Applied color coding to separate different sections visually.
- Maintained a top-to-bottom hierarchy for smooth information flow.
- Used short keywords instead of lengthy explanations to keep the map neat and readable.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, Elasticity, and Resource Management

- **Scalability** allows cloud systems to handle increasing workloads by adding or removing resources as needed.
- **Elasticity** enables resources to expand or shrink automatically based on real-time demand, improving efficiency.
- **Resource Management** ensures that computing resources are allocated, monitored, and utilized effectively.

Challenges if These Capabilities Were Absent

- Systems may become slow or unavailable during periods of high demand.
- Resources could be overused or underutilized, leading to higher operational costs.
- Performance and reliability would decrease due to inefficient resource allocation.
- The

3.Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

Justification of APIs, Web Services, and Cloud Services

- APIs enable communication between cloud applications and services, allowing seamless integration and data exchange.
- Web Services provide standardized methods for accessing cloud resources over the internet, ensuring interoperability between different systems.
- Cloud Services such as IaaS, PaaS, and SaaS deliver infrastructure, platforms, and software efficiently based on user requirements.

Contribution to Functionality and Efficiency

- Improve communication and integration between applications.
- Enable faster deployment and easy access to cloud resources.
- Reduce infrastructure management efforts and operational costs.
- Enhance system performance, flexibility, and overall reliability.

4.Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

Relationships Between Components

- Cloud service models (IaaS, PaaS, SaaS) work together to provide complete cloud solutions.
- Virtualization and web services enable efficient resource sharing and communication between applications.
- Scalability and elasticity ensure resources are adjusted according to user demand, maintaining smooth performance.
- Resource management coordinates the allocation of computing resources to improve system efficiency.

Impact on Reliability, Performance, and User Experience

- Improves system reliability by ensuring continuous service availability.
- Enhances performance through efficient resource utilization and workload distribution.
- Provides faster response times and better scalability during peak usage.
- Delivers a seamless and consistent user experience with minimal downtime.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Reflection on Developing the Concept Map

- Creating the concept map helped me understand how different cloud computing components are interconnected.
- It made complex topics easier to visualize by organizing them into a logical structure.
- Analyzing the relationships between concepts improved my understanding of cloud architecture and service models.
- The concept map also highlighted the importance of virtualization, scalability, and cloud services in building efficient systems.

Improvements for Redesign

- Add more detailed link labels to explain the relationships between concepts.
- Include real-world examples of cloud services for better understanding.
- Improve the visual layout with more balanced spacing and branch organization.
- Use additional color coding or icons to make the concept map more attractive and easier to interpret.